

Aarya Gupta

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Education

Jaypee University of Information Technology, Solan (HP), B.Tech in CSE | CGPA : 8.07/10 July 2023 – Present

- **Relevant Coursework:** Data Structures and Algorithms, DBMS (RDBMS), OOP, Operating Systems, DAA, Computer Networks

Experience

Project Intern, IIT Delhi – Delhi, India June 2025 – July 2025

- Designed and automated Cisco DNA Center REST API workflows using Postman, reducing deployment errors by 40%.
- Established Git branching and deployment workflows on Linux servers, reducing rollback time by 60%.
- Developed Bash-based Linux automation pipelines for infrastructure management, reducing recurring manual effort by 35%.

Projects

AuthForge | Authentication & Authorization Platform [Repository]

- Built a production-ready authentication platform with JWT, RBAC, refresh token rotation, and Redis token blacklisting.
- Implemented email verification, password reset, rate limiting, Redis caching, and audit logging using FastAPI.
- Containerized with Docker, integrated GitHub Actions CI, and validated functionality using 13+ Pytest test cases.

CardioVision | AI-Powered Coronary Artery Stenosis Detection [Repository]

- Developed a YOLOv8 pipeline for coronary artery stenosis detection using the ARCADE dataset and CADICA dataset.
- Achieved 94.4% mAP@0.5 and 0.91 F1-score through dataset preprocessing, augmentation, and model optimization.
- Enhanced model interpretability by integrating Grad-CAM visualizations for prediction analysis and clinical decision support.

Retail Inventory Management | REST API Backend System [Repository]

- Developed a production-grade REST API for inventory, order, category, and stock management using FastAPI and PostgreSQL.
- Implemented atomic order processing, stock restoration, and low-stock alert workflows to ensure transactional consistency.
- Applied Test-Driven Development (TDD) with 41 automated Pytest test cases, achieving a 100% test pass rate.

Skills

Languages: Python, C++ , C, JavaScript, SQL

Backend: FastAPI, Django, Express.js, REST APIs, SQLAlchemy, Pydantic, JWT, OAuth2, RBAC

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis

Testing: Pytest, Test-Driven Development (TDD), Postman, API Testing

Cloud & DevOps: Docker, Git, GitHub, GitHub Actions, Linux, AWS

AI/ML: PyTorch, YOLOv8, OpenCV, Scikit-learn, XGBoost, Pandas, NumPy

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Leadership

WebMaster - ACM JUIT Chapter

- Led a team to rebuild the ACM chapter website; instituted Git-based code review workflow reducing merge conflicts.
- Analysed real-time user feedback and web analytics to drive iterative UI/UX improvements, increasing session engagement.

Achievements and Responsibilities

- National Semi-Finalist, Flipkart GRiD 7.0; Finalist in multi inter-college coding competitions; Increased ACM hackathon participation by 150 % through strategic outreach.

Certifications

Oracle: OCI 2025 Certified AI Foundations Associate | **AWS:** Academy Graduate – Machine Learning Foundations | Academy Graduate – Cloud Foundations